Flow through a 90-degree bend

Abstract

This report aims to identify and explain the flow of a fluid through a 90-degree pipe bend modelled using STARCCM++ (a CFD tool). Two types of models are considered, 2D pipe flow and 3D pipe flow, both using a fine mesh, where the mesh gets finer nearer the walls of the models, to more accurately provide the data of how the flow interacts with the boundary. The key aspects considered when modelling these pipe sections were to quantify the pressure drop between the inlet and outlet for turbulent flows of different Reynold's number. These models are qualitatively analysed explaining the inner workings of the flow behaviour and why certain characteristics are present within the pressure and velocity fields by linking the findings to theory and literature. Furthermore, the obtained pressure drops from the simulation are then compared to analytically obtained values of pressure drop to ascertain any discrepancies between an ideal model and a simulated model. The key findings regarding this that there are large discrepancies between the 2D simulations and empirical models, ranging to a maximum variation of 387.4%, whereas the 3D show a closer correlation between the simulation and empirical models, with a variation of only up to 37.4%. This represents the closeness of the CFD model to expectations, showing that the more complex and realistic the model, the closer the values are to the empirical estimates.

Contents

1 Introduction

In fluid dynamics, pipe flow is a critical subject and has many applications in the engineering industry and elsewhere, such as chemical processing and HVAC systems.

When designing these pipes, it is imperative to minimise energy losses and ensure efficient fluid flow. However, implementing 90° pipe bends gives rise to certain unwanted flow behaviours, as they introduce complex flow patterns that straight pipe sections do not, such as eddy currents and flow separation [1]. These can greatly increase the energy lost and cause wear on the pipe walls. An indicator of energy loss in the system is a pressure loss across the bend.

This report investigates the flow dynamics in a 90° pipe bend, simulated using both two-dimensional (2D) and three-dimensional (3D) models with STARCCM++. It will explore the effects of varying Reynolds numbers on the behaviour of the fluid, with a primary focus on the effect of a 90-degree pipe bend on the pressure drop within the pipe, which is influenced by factors such as flow velocity, bend radius and pipe diameter. By comparing our simulated pressure drops with theoretical values from empirical models, we will be assessing the accuracy of computational fluid dynamics (CFD) simulations in modelling complex pipe flow.

As the fully developed flow reaches the bend of the pipe, due to centrifugal forces, the velocity and radial pressure at the walls is zero due as described by the following equation:

$$-\frac{\partial P}{\partial r} = \frac{\rho V^2}{r} (1)$$

To account for high pressure along the outer radius and low pressure on the inner radius, the fluid must accelerate on the inside of the bend, while flow deceleration and possible separation occur on the outer wall of the bend.

To determine the pressure drop across the straight sections of pipe (more specifically the inlet and outlet), the Darcy-Weisbach equation [8] can be employed:

$$\Delta P_{Loss} = f \frac{\mathrm{L}}{D} \frac{\rho V^2}{2} (2)$$

This accounts for friction factor, *f*, which is dependent on the Reynolds number and relative roughness of the pipe.

Whereas for the pressure drop along the bend section of the pipe, various empirical models were studied through a literature review, from which the two focused on within this paper are Oberg's method [2], and Spedding's method [3].

These empirical methods were mainly obtained through experimentation using wind tunnels to simulate fluid flow through bend sections in a pipe for flow of specific Reynolds numbers, which then formed the basis to find specific trends observed through the plotting of variables on graphs.

The four different Reynolds numbers analysed in this paper are: Re = 5e4, Re = 1e5, Re = 5e5, and Re = 1e6. A higher Reynolds number corresponds to a higher turbulence, so we can expect to find more eddies and vortices due to higher flow fluctuations, thus causing increased energy dissipation.

2 2-D Results

2.1 How the pipe was modelled

The 2D mesh was modelled to be as balanced as possible, in regard to simulation time and accuracy of numerical solutions. The mesh was designed with smaller control volumes nearer the boundary of the pipe and larger nearer the centre, to ensure more accuracy at the boundary of the model accounting for the viscous effects generated by the walls of the pipe. This was also due to the need to balance for the simulation time, since the smaller the control volume, the more iterations need to be run to generate data, therefore only the most important regions of flow had the smaller control volumes. The CFD was then set up with various parameters, likening the flow to be as realistic as possible when ran, but also optimising the efficiency of the simulation. For this reason, the flow was modelled using segregated flow, which is a SIMPLE discretisation scheme. This works by breaking down the continuous flow into discrete values/sub-problems, and each one is individually solved over the whole pipe, and then combined to give the whole flow solution. This is what produces the overall pressure and velocity contours in the simulation. Since the flow modelled in the pipes was turbulent, a method that would be able to accurately predict the development of the turbulent flow was required.

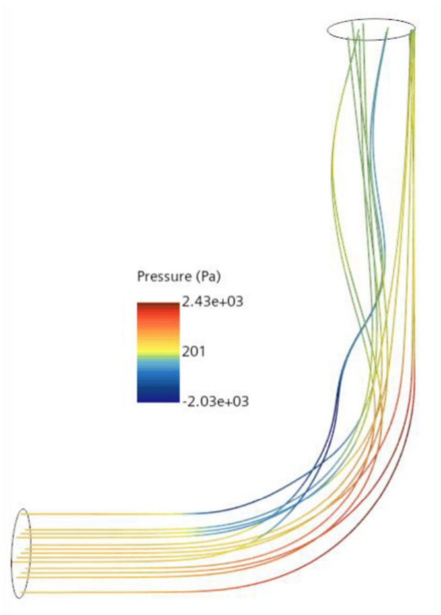

Figure 18: Pressure profile streamlines for Re = 5e5 (3D model)

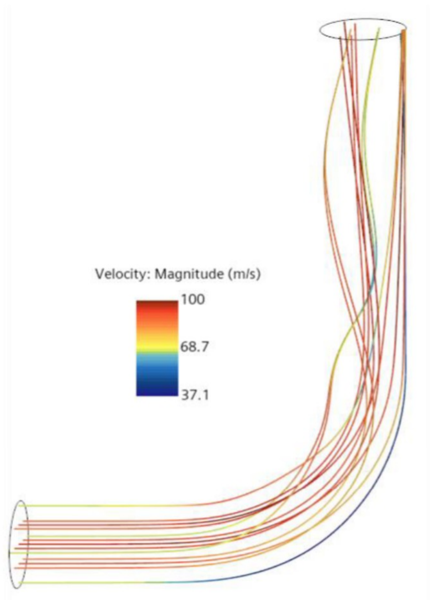

Figure 19: Velocity profile streamlines for Re = 5e5 (3D model)

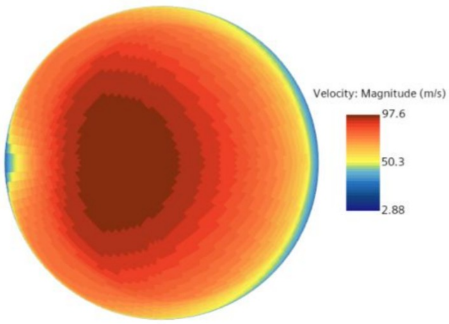

Figure 20: Velocity contours through section of pipe before bend (3D model)

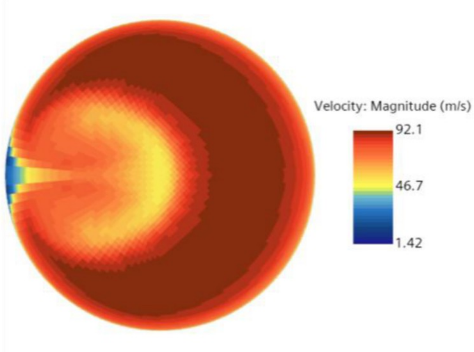

Figure 21: Velocity contours through section of pipe after bend (3D model)

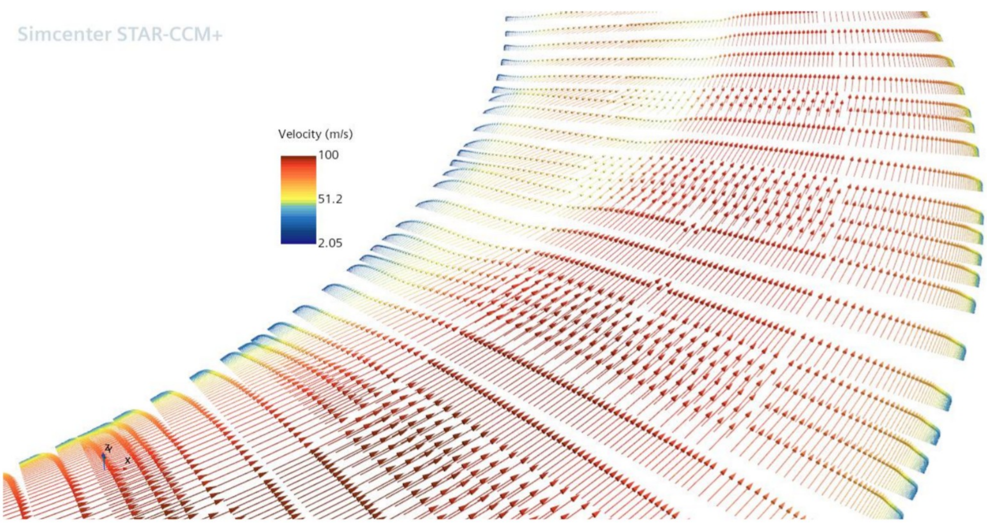

Figure 22: Velocity magnitude in vectorial representation at pipe bend (3D model)

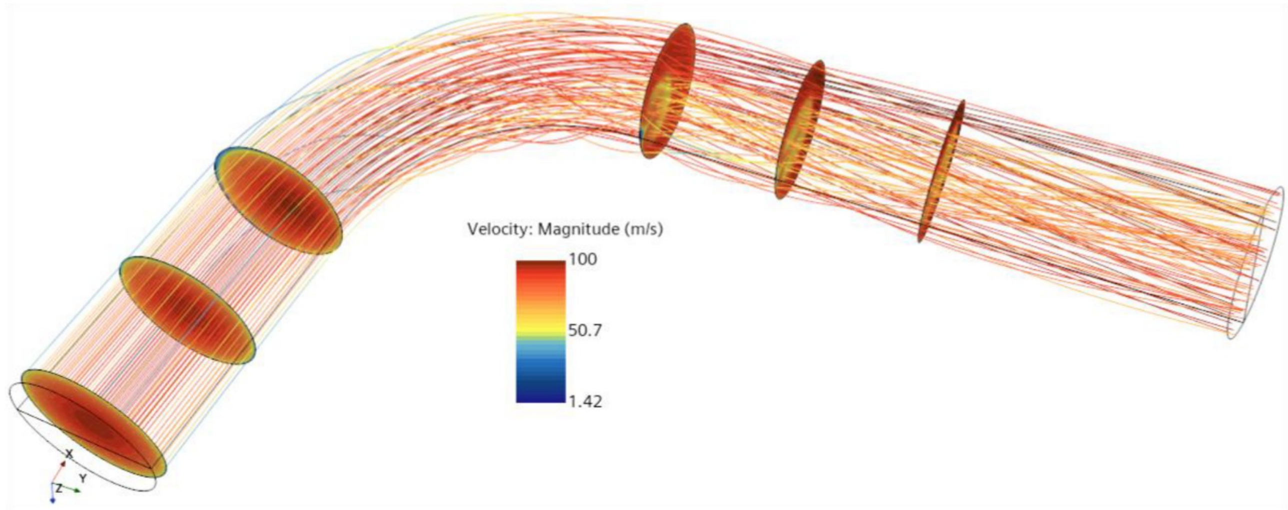

Figure 23: Velocity magnitudes shown for different sections and streamlines for Re = 5e5 (3D model)

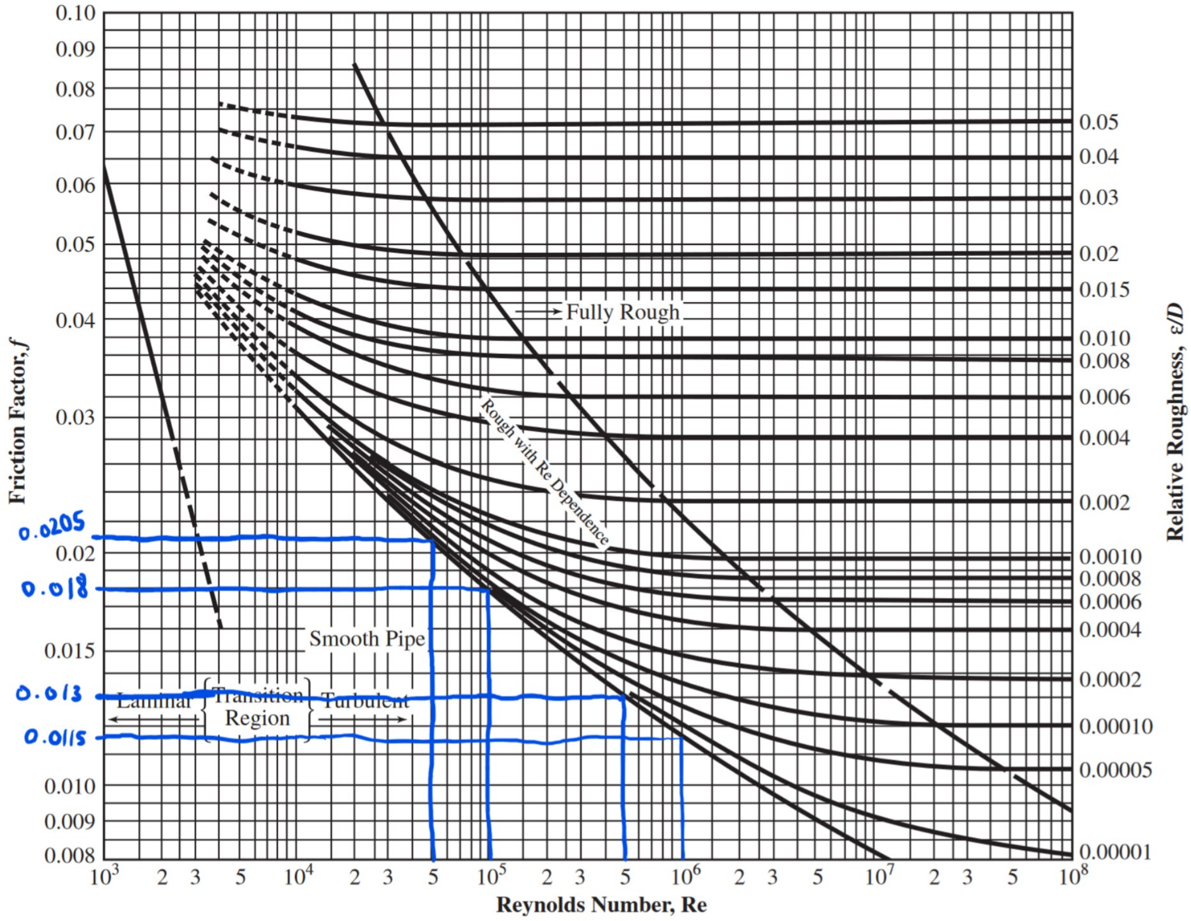

Figure 24: Moody chart showing relevant friction factors for Reynolds numbers investigated

Table 25. Bend Equivalent Length Ratio, L_e/D

Bend Type	Bend Angle	R/D	L_e/D	Bend Type	Bend Angle	R/D	L_e/D
Radius Bend	90°	1	20	Miter Bend	30°	…	8
		2	12		45°	…	15
		3	12		60°	…	25
		4	14		90°	…	60
		6	17	Close Return Bend	180°	…	50
		8	24				
		10	30				
		12	34				
		16	42				

R/D is the bend mean radius divided by pipe diameter

Figure 25: Table from Oberg machinery handbook [2]

7.2 Additional Equations/Formulae

$$\frac{I_s}{I_{s0}} = \exp\left(-\beta_s \frac{L}{D}\right)(A-1)$$

Where: β_s represents the rate at which the secondary flow intensity decreases downstream of the elbow, I_s is the swirl intensity, I_{s0} is the initial swirl intensity at the bend, L/D is the length travelled downstream of the bend over the pipe diameter.